MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3890 REAGENTS – SMBS & PEROXIDE STORAGE AND MIXING STATION
DESIGN CALCULATION OF TANK FOUNDATION

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

The Purpose of this document is to analysis & design of **3890-REA-001 SMBS MIXING TANK & 3890-TNK-001 SMBS DOSING TANK**. The dimensions and reinforcement are to be calculated. The report describes the methodology and the design parameters used in the design calculation. The analysis & design is carried out in accordance with project specifications , codes & standards.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah- Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in standard excel sheet. Stability check against Overturning & Sliding are done in the excel sheet. Base Pressure check has been done in the excel sheet. For strength calculation Bending moment & Shear Force values are calculate in the conventional method & designed in the standard excel sheet.

4. DESIGN REQUIREMENTS

4.1 Reference Codes & Books

Reference is made in this standard as indicated in STRUCTURAL ENGINEERING DESIGN CRITERIA Doc No. : MD-216-1000-EG-ST-CRT-1001-A03 , Structural Design Criteria - Process Building & Structures ITB Doc No. : MD-101-SMEM-EG-ST-CRT-1001 & Civil Design Criteria - General ITB Doc. No. MD-101-SMEM-EG-CV-CRT-1001.

The below mention codes & books are also used for this design calculation.

(i) SBC - 301
(ii) SBC - 304 CR - 2018
(iii) ACI 318M.
(iv) ACI SP-17M(09)

4.2 Materials Property

A. Density

4.2.A.(i) Density of Dry Soil : 1.9 T/m^3
4.2.A.(ii) Density of Concrete : 2.5 T/m^3
4.2.A.(iii) Density of Water : 1.0 T/m^3

B. Concrete Grade

4.2.B.(i) Foundation and Structural Members: 35 Mpa Strength of 28 days
4.2.B.(ii) Blinding Concrete below foundation bases: 15 Mpa Strength of 28 days

C. Reinforcement Grade

4.2.C.(i) Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
4.2.C.(ii) Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

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D. Grout Grade

4.2.D.(i) Grout shall be 50 mm cementitious of Strength 40 MPa.

4.3 Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm, except where welded wire fabric is used. For ties and spirals the minimum rebar size shall be 10 mm.

4.4 Concrete Clear Cover

Concrete Exposure	Member	MinimumCove (mm)
Exposed to Soil	Foundation	50
	Wall	50
	Slab on Grade	20

5. LOAD

5.1 Primary Loads

3890-REA-001 SMBS MIXING TANK & 3890-TNK-001 SMBS DOSING TANK foundation design the primary load values are taken from Drg. No. : MD-216-3890-EG-ME-GAD-1008 & MD-216-3890-EG-ME-GAD-1009 respectively.

1 DEAD LOAD (D)
2 LIVE LOAD (L)
3 SEISMIC LOAD (S)
4 WIND LOAD (W)
5 FLUID LOAD (F)

5.2 Load Combinations for Stress Calculation

LOAD CASE	LOAD COMBINATION
101	D + F
102	D + F + L
103	D + F + W
104	D + F + 0.7*E
105	D + F + 0.75*W + 0.75*L
106	D + F + 0.75*(0.7*E) + 0.75*L
107	0.6*D + W
108	0.6*D + 0.7*E

5.3 Load Combinations for Strength Calculation

LOAD CASE	LOAD COMBINATION
201	1.4*(D + F)
202	1.4*(D + F) + 1.7*L
203	1.2*D + 0.8*W
204	1.2*D + 1.6*W + 0.5*L
205	1.2*D + E + 0.5*L
206	0.9*D + 1.6*W
207	0.9*D + E

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6. SAFETY FACTOR AGAINST STABILITY

6.1 Overturning : For foundation design factor of safety against overturning is consider 2.0 for permanent load.
For foundation design factor of safety against overturning is consider 1.5 for temporary load.

6.2 Sliding : For foundation design a minimum factor of safety against sliding is consider 1.5

7. ALLOWABLE SOIL BEARING CAPACITY

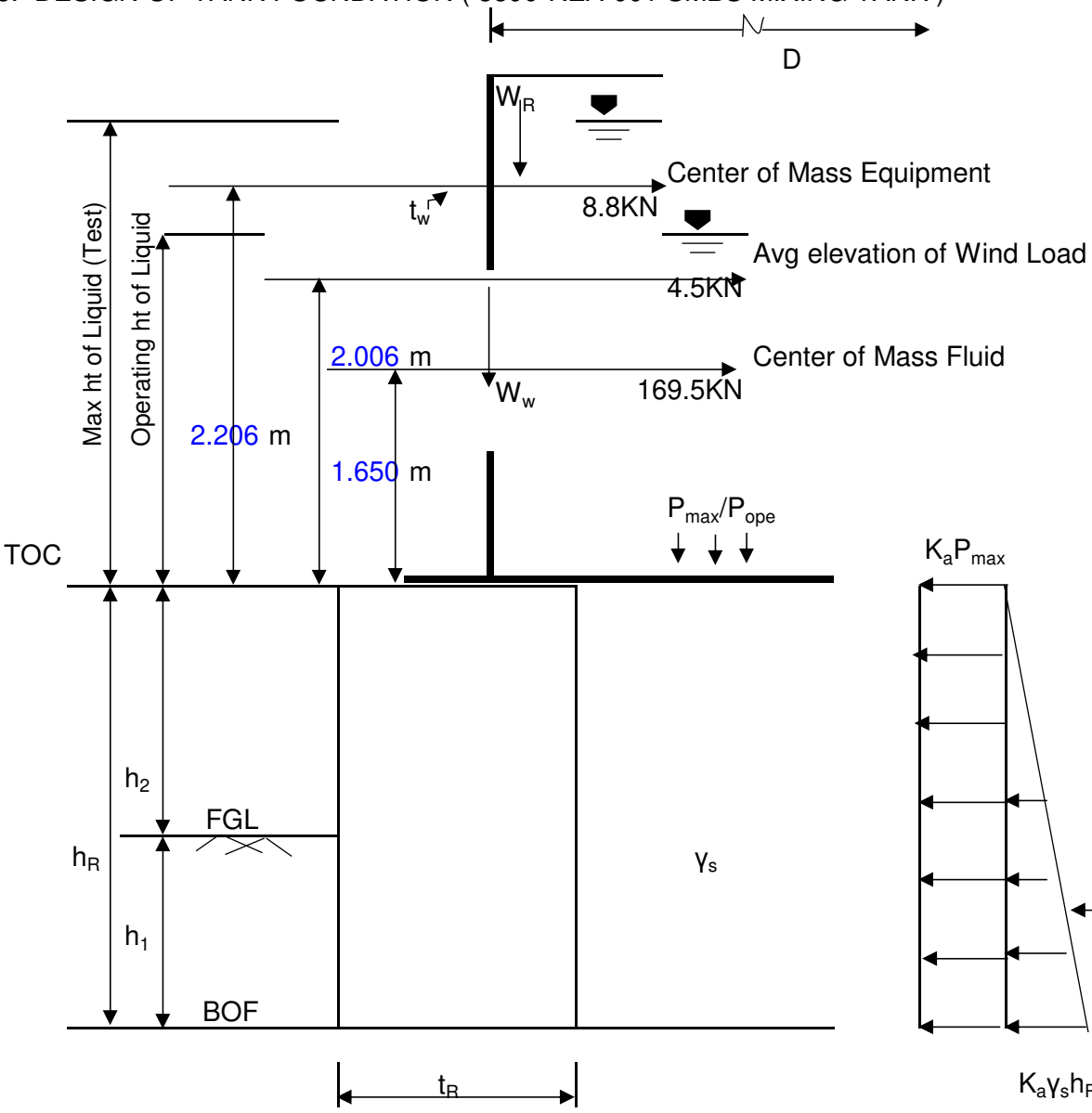
The net safe bearing capacity of soil for foundations design as obtained from page 39 of Doc. No. MD-216-0000-EM-GM-RPT-0003 is 670 KN/m2 at 1.2m depth below FGL
The neareast Bore Hole of **3890-REA-001 SMBS MIXING TANK & 3890-TNK-001 SMBS DOSING TANK** is **BH-04**.
The Net Safe Bearing Capacity of soil as per approved L&T soil test report (Table-6 of Doc. No. MD-216-3000-EG-CV-RPT-1008 -A02) is as mention below.

(i) 3m X 3m Net SBC at 1.5 below FGL is 450 KN/SQM
(ii) 4m X 4m Net SBC at 1.5 below FGL is 375 KN/SQM

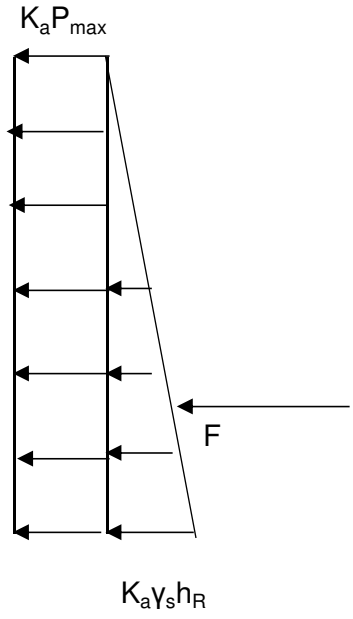
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8. DESIGN OF TANK FOUNDATION (3890-REA-001 SMBS MIXING TANK)



Typical Section Through Tank



Pressure on ring wall

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Tank Data :

Diameter of tank ,D =	2.855	m
Self weight of tank =	30.3	KN
Self weight of tank per m of circumference, W_T =	3.38	KN/m
Live Load =	32.3	KN
Live load per m of circumference, W_L =	3.6	KN/m
Maximum weight of Fluid (F) =	169.5	KN
Pressure at base of tank , P_{max} = F / A =	26.5	KN/m^2
Operating weight of Slurry ($F_{operating}$) =	0	KN
Pressure at base of tank , $P_{operating}$ = ($F_{operating}$) / A =	0	KN/m^2

Foundation Material :

Concrete, f_c' =	35	MPa
Reinforcement, f_y =	420	MPa
Unit weight of concrete, γ_c =	24	KN/m^3

Soil Data :

Unit weight of soil, γ_s =	19	KN/m^3	(Natural sand)
Allowable bearing pressure, q_{all} =	450	KN/m^2	
Coefficient of active earth pressure, K_a =	0.33		

Design Data of Ringwall :

Ringwall depth, $h_R = h_1 + h_2$ =	2.1	m
FGL to Bottom of foundation , h_1 =	1.5	m
FGL to Top of concrete , h_2 =	0.6	m
Ringwall thickness, t_R =	450	mm
Inner dia of ringwall, D_i =	2.455	m
Outer dia of ringwall, D_o =	3.355	m
$S_{xx} = \pi(D_o^4-D_i^4)/32D_o$ =	2.64	m^3

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Check of soil bearing pressure (Max weight of Slurry)

Wind force = 4.5 KN acting at a height of 4.106 m from BOF

M_{OT} = Overturning moment = 18.5 KN-m

M_{OT} / S_{xx} = 6.988 KN/m^2

Load Combination : 102. D + F + L

$q_{RF} = (W_{T+} W_L)/t_R + P_{max} + h_R\gamma_c$ = 92.39 KN/m^2 **< 450KN/m2, O.K.**

Load Combination : 103. D + F ± W

$q_{max} = (W_T)/t_R + P_{max} + h_R\gamma_c + M_{OT}/S_{xx}$ = 91.37 KN/m^2 **< 450KN/m2, O.K.**

$q_{min} = (W_T)/t_R + P_{max} + h_R\gamma_c - M_{OT}/S_{xx}$ = 77.4 KN/m^2 **No (-)ve pressure**

Load Combination : 105. D + F ± 0.75*W + 0.75*L

$q_{max} = (W_T+0.75*W_L)/t_R + P_{max} + h_R\gamma_c + 0.75*M_{OT}/S_{xx}$ = 95.63 KN/m^2 **< 450KN/m2, O.K.**

$q_{min} = (W_T+0.75*W_L)/t_R + P_{max} + h_R\gamma_c - 0.75*M_{OT}/S_{xx}$ = 85.15 KN/m^2 **No (-)ve pressure**

Check of soil bearing pressure (Empty Tank): For wind Load

Soil bearing pressure under ringwall :

Load Combination : 107. 0.6*D ± W

$q_{RE} = (W_T)/t_R + h_R\gamma_c$ = 57.9 KN/m^2

$q_{max} = 0.6*q_{RE} + M_{OT}/S_{xx}$ = 41.7 KN/m^2 **< 450KN/m2, O.K.**

$q_{min} = 0.6*q_{RE} - M_{OT}/S_{xx}$ = 27.76 KN/m^2 **No (-)ve pressure**

Check for overturning (Empty Tank): For wind Load

M_{OT} = Overturning moment = 18.5 KN-m

M_{OR} = Restoring moment = (30.3+PI()*2.855*0.45*2.1*24)*2.855/2 = 334 KN-m

FOS = M_{OR} / M_{OT} = 333.64 / 18.48 = 18.05 **> 1.5, Hence O.K.**

Check for soil bearing pressure (Operating weight of Fluid):

Seismic force at center of mass equipment = 8.8 KN

acting at a height of 4.306 m from BOF

M_{OT} = Overturning moment = 37.89 KN-m

M_{OT} / S_{xx} = 14.33 KN/m^2

Load Combination : 101. D + F

$q_{RF} = (W_T)/t_R + F + h_R\gamma_c$ = 84.38 KN/m^2 **< 450KN/m2, O.K.**

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Load Combination : 107. D + F ± 0.7*E

$q_{max} = q_{RF} + 0.7*M_{OT}/S_{xx}$ = 94.41 KN/m^2 **< 450KN/m2, O.K.**

$q_{min} = q_{RF} - 0.7*M_{OT}/S_{xx}$ = 74.35 KN/m^2 **No (-)ve pressure**

Load Combination : 108. 0.6*D ± 0.7*E

$q_{max} = 0.6*(W_T)/t_R + 0.6*h_R\gamma_c + 0.7*M_{OT}/S_{xx}$ = 44.77 KN/m^2 **< 450KN/m2, O.K.**

$q_{max} = 0.6*(W_T)/t_R + 0.6*h_R\gamma_c - 0.7*M_{OT}/S_{xx}$ = 24.71 KN/m^2 **No (-)ve pressure**

Check for Overturning (Empty Tank): For Seismic Load

M_{OT} = Overturning moment = 37.9 KN-m

M_{OR} = Restoring moment = (30.3+PI()*2.855*0.45*2.1*24)*2.855/2 = 334 KN-m

FOS = M_{OR} / M_{OT} = 333.64 / 37.8928 = 8.80 **> 1.5, Hence O.K.**

Ringwall Reinforcement :

Radial outward force on ringwall, $R_{F1} = K_a h_R(P_{max} + \gamma_s h_R/2)$ = 32.2 KN/m

For Hydrotest :

Radial outward force on ringwall, R_{F2}= $K_a h_R(\gamma_w h_L + \gamma_s h_R)$ = 34.3 Kn/m

Hoop tension in ringwall, T = 1/2*Max(R_{F1},R_{F2})*D = 48.92 KN

Factored design tension, T_u= 68.49 KN FOS = 1.4

$A_{st,reqd} = T_u/\phi f_y$ = 181.2 mm^2 ϕ = 0.9

$A_{st,min}$ = 0.002* $h_R t_R$ = 1890 mm^2

$A_{st,reqd}$ (per face of ring wall)= 945 mm^2

Required : 12 dia bar @ 251.4 c/c (per face)

Provide : 12 dia bar @ 200 c/c (per face)

$A_{st,\ provided}$(per face) = 1188 mm^2 **O.K.**

Vertical Reinforcement :

$A_{st,min}$ = 0.0012* t_R = 540 mm^2/m

$A_{st,reqd}$ (per face of ring wall)= 270 mm^2

Required : 10 dia bar @ 291.1 c/c (per face)

Provide : 10 dia bar @ 200 c/c (per face)

$A_{st,\ provided}$(per face) = 392.9 mm^2 **O.K.**

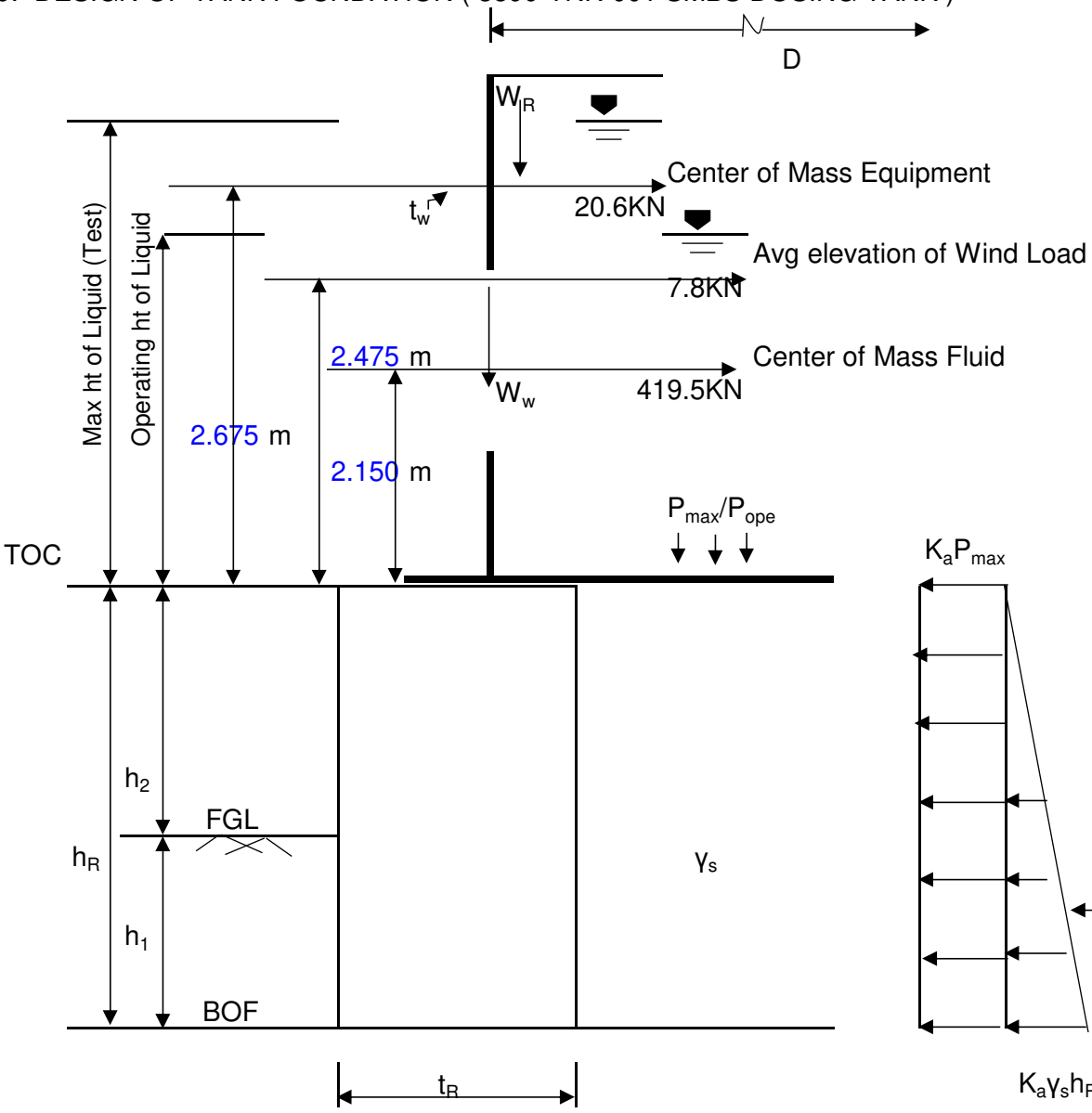
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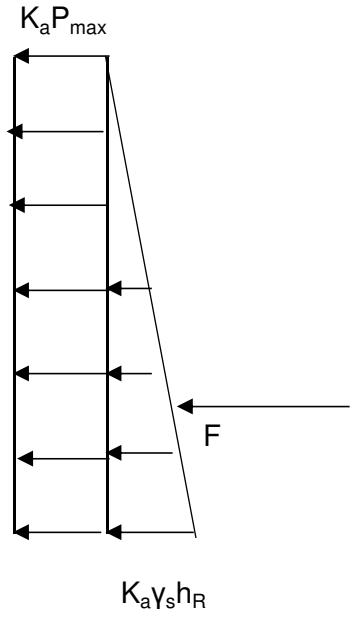
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9. DESIGN OF TANK FOUNDATION (3890-TNK-001 SMBS DOSING TANK)



Typical Section Through Tank



Pressure on ring wall

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Tank Data :

Diameter of tank ,D =	3.810	m
Self weight of tank =	40.3	KN
Self weight of tank per m of circumference, W_T =	3.37	KN/m
Live Load =	57.3	KN
Live load per m of circumference, W_L =	4.79	KN/m
Maximum weight of Fluid (F) =	419.5	KN
Pressure at base of tank , P_{max} = F / A =	36.8	KN/m^2

Foundation Material :

Concrete, f_c' =	35	MPa
Reinforcement, f_y =	420	MPa
Unit weight of concrete, γ_c =	24	KN/m^3

Soil Data :

Unit weight of soil, γ_s =	19	KN/m^3	(Natural sand)
Allowable bearing pressure, q_{all} =	375	KN/m^2	
Coefficient of active earth pressure, K_a =	0.33		

Design Data of Ringwall :

Ringwall depth, $h_R = h_1 + h_2$ =	2.1	m
FGL to Bottom of foundation , h_1 =	1.5	m
FGL to Top of concrete , h_2 =	0.6	m
Ringwall thickness, t_R =	450	mm
Inner dia of ringwall, D_i =	3.41	m
Outer dia of ringwall, D_o =	4.31	m
$S_{xx} = \pi(D_o^4-D_i^4)/32D_o$ =	4.78	m^3

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MA'ADEN معادن Saudi Arabian Mining Company شركة التعدين العربية السعودية		Outotec / LARSEN & TOUBRO
MD-216-3890-EG-CV-CAL-1436	Mansourah - Massarah Gold Project	Page: of
900707_GAU01_AA04_CED01_ 38902_OU100688671_R2	CALCULATION REAGENTS–SMBS & PEROXIDE STORAGE AND MIXING STATION DESIGN CALCULATION OF TANK FOUNDATION	Date: 30.11.2023
O19045_450_C_K_24172_001_R2	AS BUILT	Rev: D01

Check of soil bearing pressure (Max weight of Slurry)

Wind force = 7.8 KN acting at a height of 4.575 m from BOF

M_{OT} = Overturning moment = 35.7 KN-m

M_{OT} / S_{xx} = 7.466 KN/m^2

Load Combination : 102. D + F + L

$q_{RF} = (W_{T+} W_L)/t_R + P_{max} + h_R\gamma_c$ = 105.3 KN/m^2 **< 375KN/m2, O.K.**

Load Combination : 103. D + F ± W

$q_{max} = (W_T)/t_R + P_{max} + h_R\gamma_c + M_{OT}/S_{xx}$ = 102.1 KN/m^2 **< 375KN/m2, O.K.**

$q_{min} = (W_T)/t_R + P_{max} + h_R\gamma_c - M_{OT}/S_{xx}$ = 87.21 KN/m^2 **No (-)ve pressure**

Load Combination : 105. D + F ± 0.75*W + 0.75*L

$q_{max} = (W_T+0.75*W_L)/t_R + P_{max} + h_R\gamma_c + 0.75*M_{OT}/S_{xx}$ = 108.3 KN/m^2 **< 375KN/m2, O.K.**

$q_{min} = (W_T+0.75*W_L)/t_R + P_{max} + h_R\gamma_c - 0.75*M_{OT}/S_{xx}$ = 97.06 KN/m^2 **No (-)ve pressure**

Check of soil bearing pressure (Empty Tank): For wind Load

Soil bearing pressure under ringwall :

Load Combination : 107. 0.6*D ± W

$q_{RE} = (W_T)/t_R + h_R\gamma_c$ = 57.9 KN/m^2

$q_{max} = 0.6*q_{RE} + M_{OT}/S_{xx}$ = 42.2 KN/m^2 **< 375KN/m2, O.K.**

$q_{min} = 0.6*q_{RE} - M_{OT}/S_{xx}$ = 27.26 KN/m^2 **No (-)ve pressure**

Check for overturning (Empty Tank): For wind Load

M_{OT} = Overturning moment = 35.7 KN-m

M_{OR} = Restoring moment = (40.3+PI()*3.81*0.45*2.1*24)*3.81/2 = 594 KN-m

FOS = M_{OR} / M_{OT} = 593.92 / 35.69 = 16.64 **> 1.5, Hence O.K.**

Check for soil bearing pressure (Operating weight of Fluid):

Seismic force at center of mass equipment = 20.6 KN

acting at a height of 4.775 m from BOF

M_{OT} = Overturning moment = 98.37 KN-m

M_{OT} / S_{xx} = 20.58 KN/m^2

Load Combination : 101. D + F

$q_{RF} = (W_T)/t_R + F + h_R\gamma_c$ = 94.68 KN/m^2 **< 375KN/m2, O.K.**

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MD-216-3890-EG-CV-CAL-1436	Mansourah - Massarah Gold Project	Page: of
900707_GAU01_AA04_CED01_ 38902_OU100688671_R2	CALCULATION REAGENTS–SMBS & PEROXIDE STORAGE AND MIXING STATION DESIGN CALCULATION OF TANK FOUNDATION	Date: 30.11.2023
O19045_450_C_K_24172_001_R2	AS BUILT	Rev: D01

Load Combination : 107. D + F ± 0.7*E

$q_{max} = q_{RF} + 0.7*M_{OT}/S_{xx}$ = 109.1 KN/m^2 **< 375KN/m2, O.K.**

$q_{min} = q_{RF} - 0.7*M_{OT}/S_{xx}$ = 80.27 KN/m^2 **No (-)ve pressure**

Load Combination : 108. 0.6*D ± 0.7*E

$q_{max} = 0.6*(W_T)/t_R + 0.6*h_R\gamma_c + 0.7*M_{OT}/S_{xx}$ = 49.13 KN/m^2 **< 375KN/m2, O.K.**

$q_{max} = 0.6*(W_T)/t_R + 0.6*h_R\gamma_c - 0.7*M_{OT}/S_{xx}$ = 20.33 KN/m^2 **No (-)ve pressure**

Check for Overturning (Empty Tank): For Seismic Load

M_{OT} = Overturning moment = 98.4 KN-m

M_{OR} = Restoring moment = (40.3+PI()*3.81*0.45*2.1*24)*3.81/2 = 594 KN-m

FOS = M_{OR} / M_{OT} = 593.92 / 98.365 = 6.04 **> 1.5, Hence O.K.**

Ringwall Reinforcement :

Radial outward force on ringwall, $R_{F1} = K_a h_R(P_{max} + \gamma_s h_R/2)$ = 39.3 KN/m

For Hydrotest :

Radial outward force on ringwall, R_{F2}= $K_a h_R(\gamma_w h_L + \gamma_s h_R)$ = 34.3 Kn/m

Hoop tension in ringwall, T = 1/2*Max(R_{F1},R_{F2})*D = 74.91 KN

Factored design tension, T_u= 104.9 KN FOS = 1.4

$A_{st,reqd} = T_u/\phi f_y$ = 277.5 mm^2 ϕ = 0.9

$A_{st,min}$ = 0.002* $h_R t_R$ = 1890 mm^2

$A_{st,reqd}$ (per face of ring wall)= 945 mm^2

Required : 12 dia bar @ 251.4 c/c (per face)

Provide : 12 dia bar @ 200 c/c (per face)

$A_{st, provided}$(per face) = 1188 mm^2 **O.K.**

Vertical Reinforcement :

$A_{st,min}$ = 0.0012* t_R = 540 mm^2/m

$A_{st,reqd}$ (per face of ring wall)= 270 mm^2

Required : 10 dia bar @ 291.1 c/c (per face)

Provide : 10 dia bar @ 200 c/c (per face)

$A_{st, provided}$(per face) = 392.9 mm^2 **O.K.**

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MD-216-3890-EG-CV-CAL-1436	Mansourah Massarah Gold Project	Page	OF
900707_GAU01_AA04_CED01_ 38902_OU100688671_R2	CALCULATION REAGENTS–SMBS & PEROXIDE STORAGE AND MIXING STATION DESIGN CALCULATION OF TANK FOUNDATION	Date	30.11.2023
O19045_450_C_K_24172_001_R2	AS BUILT	Rev.	D01

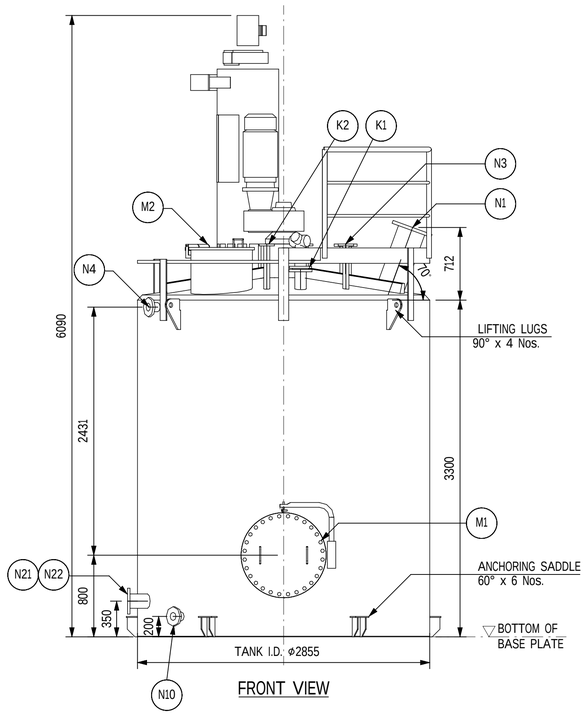
ATTACHMENT - 1 : Mechanical Input Drawing.

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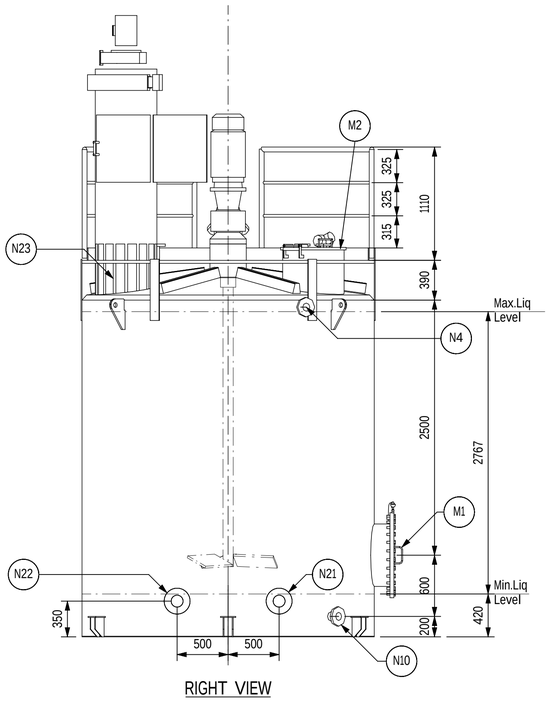
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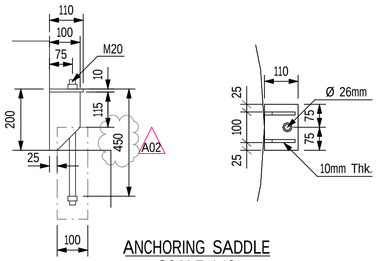
DRAWING NUMBER MD-216-3890-EG-ME-GAD-1008 REV A02



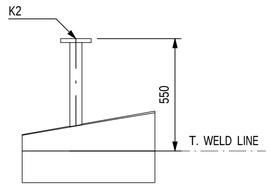
FRONT VIEW



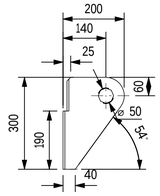
RIGHT VIEW



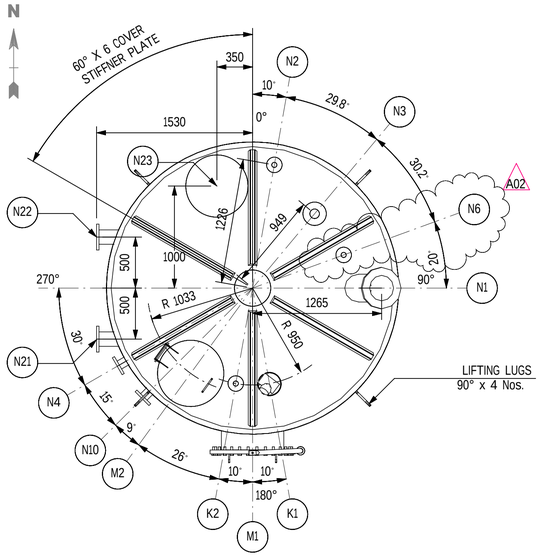
ANCHORING SADDLE
SCALE 1:10



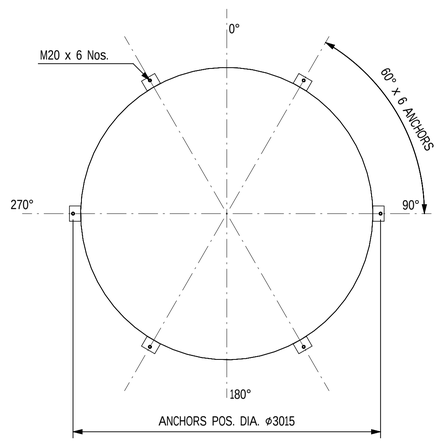
K2 NOZZLE DETAIL
NOT IN SCALE



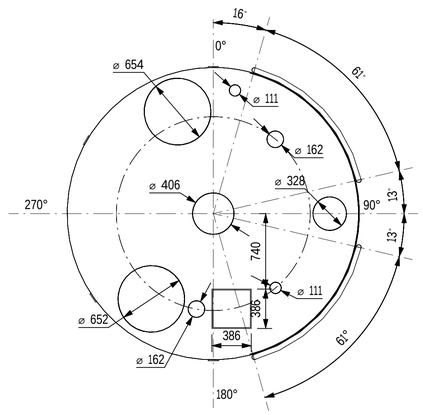
LIFTING LUGS
NOT IN SCALE



TANK TOP VIEW



ANCHORING



PLATFORM

DESIGN DATA

EQUIPMENT TAG NOs. / NAME PLATE	3890-REA-001
DESCRIPTION	SMBS MIXING TANK
DESIGN CODE	API 650
DESIGN TEMPERATURE	+0-60°C
DESIGN PRESSURE	90.97 kPa Absolute
OPERATING PRESSURE	ATM
WIND PRESSURE/ CODE	Site C / 43 m/s / SBC 301,2007 Kz=ELEV, Kzt=1.0, Kd=0.95
SHELL THICKNESS	5 mm
BOTTOM THICKNESS	6 mm
CORROSION ALLOWANCE: SHELL INSIDE	0.0016 mm
CORROSION ALLOWANCE: BOTTOM	0.0016 mm
TESTING	AS PER APPROVED Q.A.P
RADIOGRAPHY	AS PER APPROVED Q.A.P
JOINT EFFICIENCY	AS PER APPROVED Q.A.P
LOCATION	OUTDOOR
MAIN DIMENSIONS	Dia 2865 x 3300
CAPACITY: FULL VOLUME	18.9 m³
CAPACITY: LIVE VOLUME	17.7 m³
SEISMIC DESIGN /CODE	Site B / SBC 301, 2007 Ss=0.1g / S1=0.02g / Fa=1.0 / Fv=1.0 Sds=0.067g / Sd1=0.013g / TL=4s

LOAD DATA

TOTAL WEIGHT (FULL)	19844 kg
TOTAL WEIGHT (EMPTY)	2740 kg
RUBBER LINING WEIGHT	
TANK WEIGHT	2740 kg
DETAIL WEIGHTS AND MATERIAL THICKNESSES ACCORDING TO VENDOR DETAIL ENGINEERING	

MATERIAL OF CONSTRUCTION

PLATE - BOTTOM/SHELL/LIFTING LUGS	1.4301/1.4307
STRUCTURALS	EN10025-2 S235JR
NOZZLE	EN 1.4301
NOZZLE FLANGES	ASME B16.5 SO RF
SURFACE TREATMENT	WELD JOINT PICKLING
RUBBER LINING (INNER SURFACES)	-

NOZZLES TABLE

No.	SIZE	RATING	DIRECTION	PROJECTION	OFFSET	ELEVATION	STANDARD	SERVICE	QUANTITY	REMARKS
N1	10" (DN250)	CLASS 150	90°	-	-	4012	ASME B16.5	INLET	1	
N2	2" (DN50)	CLASS 150	10°	-	-	3850	ASME B16.5	RAW WATER INLET	1	
N3	4" (DN100)	CLASS 150	39.8°	-	-	3850	ASME B16.5	VENT	1	
N4	3" (DN80)	CLASS 150	240°	1533	-	3231	ASME B16.5	OVERFLOW	1	
N6	2" (DN50)	CLASS 150	70° A02	-	-	3850	ASME B16.5	SPARE A02	1	
N10	3" (DN80)	CLASS 150	225°	1533	-	200	ASME B16.5	DRAIN	1	
N21	5" (DN125)	CLASS 150	270°	1530	500	350	ASME B16.5	OUTLET	1	
N22	5" (DN125)	CLASS 150	270°	1530	500	350	ASME B16.5	OUTLET	1	
N23	24" (DN600)		270°	3500	500		ASME B16.5	DUST FILTER	1	
K1	4" (DN100)	CLASS 150	170°	-	-	3602	ASME B16.5	LEVEL TRANSMITTER	1	
K2	2" (DN50)	CLASS 150	190°	-	-	3850	ASME B16.5	LEVEL SWITCH	1	
M1	24"(DN600)	CLASS 150	180°	1603	-	800	OUTOTEC	MANHOLE	1	
M2	24"(DN600)		216°	-	-	3817	-	MANHOLE	1	

LOAD TABLE

3890-REA-001	Dead Load	Seismic	Wind	Live Load	Fluid Load
Shear Fxy (kN)	0.0	8.8	4.5	0.0	0.0
Axial Fz (kN)	30.3	9.7	5.2	32.3	169.5
Mx (kNm)	0.0	16.6	9.9	0.0	0.1
My (kNm)	0.0	16.6	9.9	0.0	0.1
Mz (kNm)	0.0	0.0	0.0	0.0	0.4

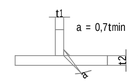
NOTE:
1. REPORTED WIND LOADS ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC301-2007. CIVIL DESIGNER TO APPLY COMBINATION FACTOR OF 1.6 FOR STRENGHT DESIGN
2. REPORTED SEISMIC LOAD IS A PRIMARY LOAD, DERIVED IN ACCORDANCE WITH THE SITE SPECIFICATIONS AND THE REQUIREMENTS OF SBC301 AND IS SUITABLE FOR COMBINATION IN ACCORDANCE WITH SBC301 FOR EITHER STRENGHT DESIGN OR ALLOWABLE STRESS DESIGN. COMBINATION FACTOR FOR STRENGHT DESIGN SHALL BE 1.0, HOWEVER IF THE LOADS ARE TO BE USED FOR ALLOWABLE STRESS DESIGN METHOD THEN THE COMBINATION FACTOR WOULD BE 0.7.
3. ALL DIMENSIONS ARE IN "mm" UNLESS OTHERWISE STATED.
4. FINAL BOLT LENGTH TO BE DEFINED ON CIVIL FOUNDATIONS DRAWINGS.

WELDING GENERAL NOTES

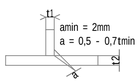
ALL WELDING SEAMS , MUST BE WELDED CONTINOUSLY UNLESS OTHERWISE SPECIFIED.

TYPICAL FILLET WELD THICKNESS:

ONE SIDE WELDING :



GENERAL :



EXECUTE WELDING WORK ACCORDING TO DIN EN ISO 5817:
- FOR BUTT WELDS, QUALITY LEVEL C,
- FOR FILLET WELDS, QUALITY LEVEL B.

GENERAL TOLERANCE FOR WELDED CONSTRUTION ACCORDING TO EN ISO 13920 - AE.

GENERAL TOLERANCE acc. to ISO 2768 mH

MM	Machined MM	none Machined MM	Straightness Flatness MM
<= 6	+ 0.1	+ 0.5	0.008
> 6 - 30	+ 0.2	+ 1.0	0.020
> 30 - 120	+ 0.3	+ 1.5	0.040
> 120 - 400	+ 0.5	+ 2.0	0.080
> 400 - 1000	+ 0.8	+ 3.0	0.150
> 1000 - 2000	+ 1.2	+ 4.0	0.200
> 2000 - 4000	+ 2.0	+ 6.0	
> 4000	+ 3.0	+ 8.0	

⫽ CONCENTRIC and AXIAL RUN TOLERANCE 0.1 MM

⎯ SYMMETRIC TOLERANCE 0.5 MM

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A02	21-07-20	ISSUED FOR APPROVAL	LG	RM	SC	RL	
A01	03-06-20	ISSUED FOR APPROVAL	LG	RM	SC	RL	

DOCUMENT NUMBER | REFERENCE DOCUMENTS

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER: 019045-450-U-G-23056-001 REVISION: 1

OUTOTEC DRAWING NUMBER: 900707 _GAU01 _AA04_MTA01 _38901 DOCUMENT ID: OU100662223 REVISION: 1

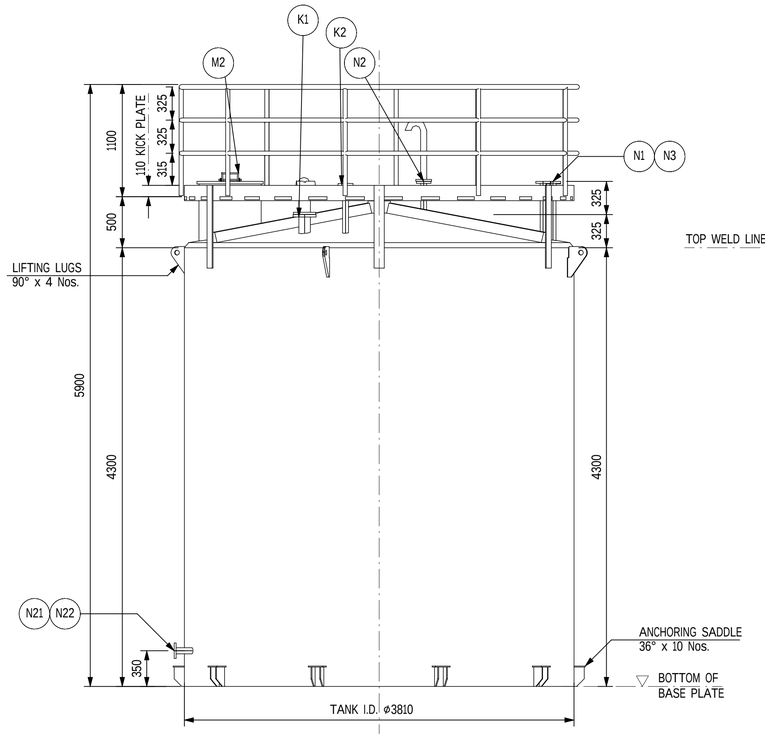
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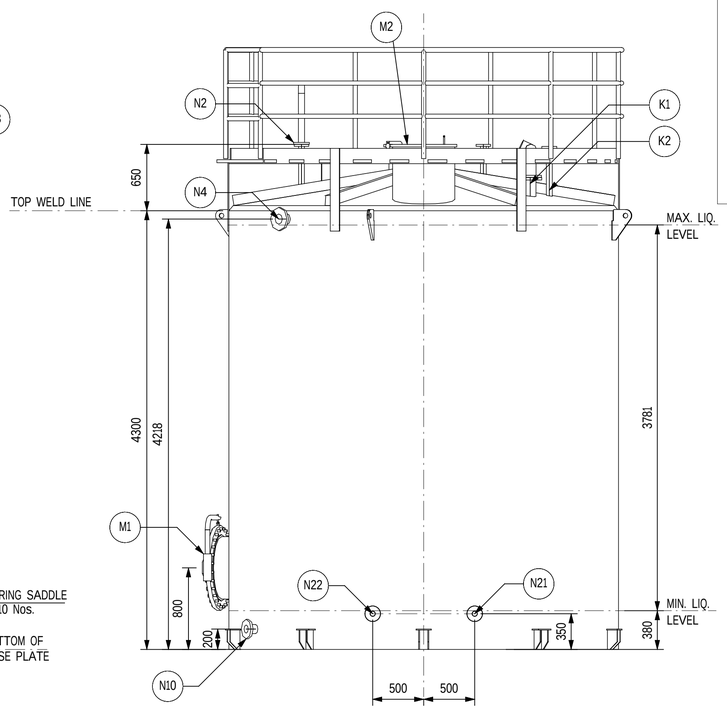
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Saudi Arabian Mining Company شركة التعدين العربية السعودية
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DRAWN	LokGav	21-07-20	MANSOURAH-MASSARAH GOLD PROJECT
CHECKED	Rohmul	21-07-20	OUTLINE DRAWING
ENGINEER	SouCha	21-07-20	SMBS & PER. STORG. & MIXING STN
APPROVED	RicLeh	21-07-20	3890-REA-001
EMPLOYER			
SCALE 1:30		A1	DRAWING NUMBER: MD-216-3890-EG-ME-GAD-1008 REV: A02

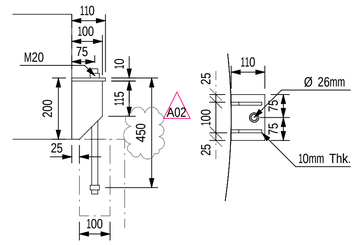
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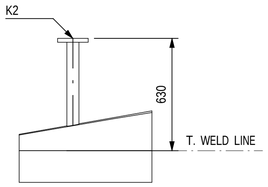
FRONT VIEW



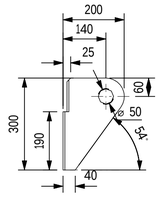
LEFT VIEW



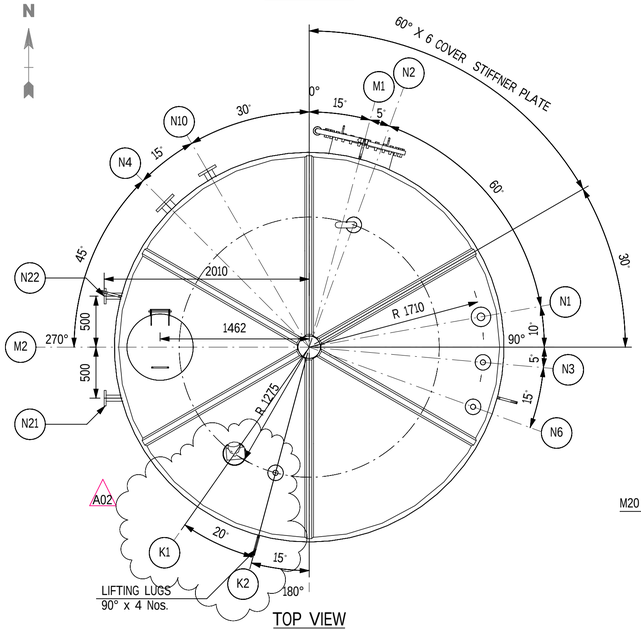
ANCHORING SADDLE
SCALE 1:10



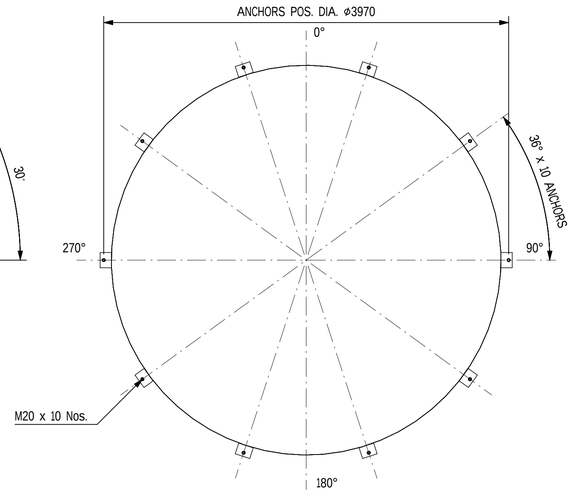
K2 NOZZLE DETAIL
NOT IN SCALE



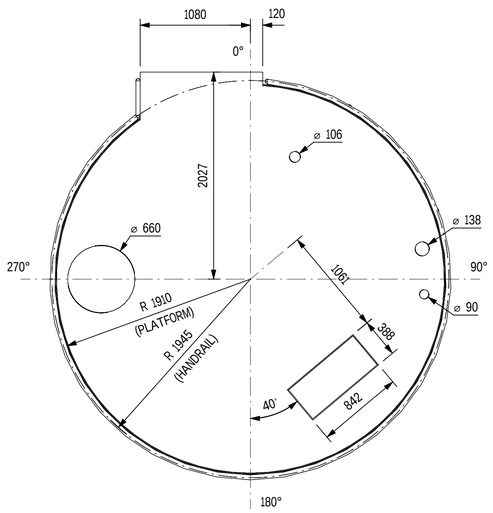
LIFTING LUGS
NOT IN SCALE



TOP VIEW



ANCHORING LOCATION



PLATFORM

DESIGN DATA

EQUIPMENT TAG NOs. / NAME PLATE	3890-TNK-001
DESCRIPTION	SMBS DOSING TANK
DESIGN CODE	API 650
DESIGN TEMPERATURE	+0-60°C
DESIGN PRESSURE	90.97 kPa Absolute
OPERATING PRESSURE	ATM
WIND PRESSURE/ CODE	Site C / 43 m/s / SBC 301,2007 Kz=ELEV, Kzt=1.0, Kd=0.95
SHELL THICKNESS	5 mm And 6 mm (from down 1800 mm)
BOTTOM THICKNESS	6 mm
CORROSION ALLOWANCE: SHELL INSIDE	0.0016 mm
CORROSION ALLOWANCE: BOTTOM	0.0016 mm
TESTING	AS PER APPROVED Q.A.P
RADIOGRAPHY	AS PER APPROVED Q.A.P
JOINT EFFICIENCY	AS PER APPROVED Q.A.P
LOCATION	OUTDOOR
MAIN DIMENSIONS	Dia 3820x4300 mm
CAPACITY: FULL VOLUME	44.7 m³
CAPACITY: LIVE VOLUME	43.1 m³
SEISMIC DESIGN /CODE	Site B / SBC 301, 2007 Ss=0.1g / S1=0.02g / Fa=1.0 / Fv=1.0 Sds=0.067g / Sd1=0.013g / TL=4s

LOAD DATA

TOTAL WEIGHT (FULL)	46870 kg
TOTAL WEIGHT (EMPTY)	4110 kg
RUBBER LINING WEIGHT	
TANK WEIGHT	4110 kg
DETAIL WEIGHTS AND MATERIAL THICKNESSES ACCORDING TO VENDOR DETAIL ENGINEERING	

MATERIAL OF CONSTRUCTION

PLATE - BOTTOM/SHELL/LIFTING LUGS	1.4306/1.4307
STRUCTURALS	EN10025- S235JR
NOZZLE	EN 1.4306
NOZZLE FLANGES	ASME B16.5 SO RF
SURFACE TREATMENT	ASME B16.5 SO RF
RUBBER LINING (INNER SURFACES)	weld joint pickling

NOZZLES TABLE

No.	SIZE	RATING	DIRECTION	PROJECTION	OFFSET	ELEVATION	STANDARD	SERVICE	QUANTITY	REMARKS
N1	3" (DN80)	CLASS 150	80°	-	-	4950	ASME B16.5	INLET	1	
N2	2" (DN50)	CLASS 150	20°	-	-	4950	ASME B16.5	VENT	1	
N3	2" (DN50)	CLASS 150	95°	-	-	4950	ASME B16.5	INLET	1	
N4	4" (DN100)	CLASS 150	315°	2008	-	4218	ASME B16.5	OVERFLOW	1	
N6	2" (DN50)	CLASS 150	110°	-	-	4950	ASME B16.5	SPARE A02	1	
N10	3" (DN80)	CLASS 150	330°	2008	-	200	ASME B16.5	DRAIN	1	
N21	3" (DN50)	CLASS 150	270°	2010	500	350	ASME B16.5	OUTLET	1	
N22	3" (DN50)	CLASS 150	270°	2010	500	350	ASME B16.5	OUTLET	1	
K1	4" (DN100)	CLASS 150	215°	-	-	4625	ASME B16.5	LEVEL TRANSMITTER	1	
K2	2" (DN50)	CLASS 150	195°	-	-	4930	ASME B16.5	LEVEL SWITCH	1	
M1	24"(DN600)	CLASS 150	18°	1650	-	800	OUTOTEC	MANHOLE	1	
M2	24"(DN600)		270°	-	-	4950	-	MANHOLE	1	

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LOAD TABLE

3890-TNK-001	Dead Load	Seismic	Wind	Live Load	Fluid Load
Shear Fxy (kN)	0,0	20.6	7.8	0,0	0,0
Axial Fz (kN)	40.3	22.9	9.2	57.3	419.5
Mx (kNm)	0,0	50.0	22.4	0,0	0,0
My (kNm)	0,0	50.0	22.4	0,0	0,0
Mz (kNm)	0,0	0,0	0,0	0,0	0,0

NOTE:
1. REPORTED WIND LOADS ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC301-2007. CIVIL DESIGNER TO APPLY COMBINATION FACTOR OF 1.6 FOR STRENGHT DESIGN
2. REPORTED SEISMIC LOAD IS A PRIMARY LOAD, DERIVED IN ACCORDANCE WITH THE SITE SPECIFICATIONS AND THE REQUIREMENTS OF SBC301 AND IS SUITABLE FOR COMBINATION IN ACCORDANCE WITH SBC301 FOR EITHER STRENGHT DESIGN OR ALLOWABLE STRESS DESIGN. COMBINATION FACTOR FOR STRENGHT DESIGN SHALL BE 1.0, HOWEVER IF THE LOADS ARE TO BE USED FOR ALLOWABLE STRESS DESIGN METHOD THEN THE COMBINATION FACTOR WOULD BE 0.7.
3. ALL DIMENSIONS ARE IN "mm" UNLESS OTHERWISE STATED.
4. FINAL BOLT LENGTH TO BE DEFINED ON CIVIL FOUNDATIONS DRAWINGS.

WELDING GENERAL NOTES

ALL WELDING SEAMS , MUST BE WELDED CONTINOUSLY UNLESS OTHERWISE SPECIFIED.

TYPICAL FILLET WELD THICKNESS:

ONE SIDE WELDING : GENERAL :



EXECUTE WELDING WORK ACCORDING TO DIN EN ISO 5817:
- FOR BUTT WELDS, QUALITY LEVEL C,
- FOR FILLET WELDS, QUALITY LEVEL B,

GENERAL TOLERANCE FOR WELDED CONSTRUTION ACCORDING TO EN ISO 13920 - AE.

GENERAL TOLERANCE acc. to ISO 2768 mH

MM	Machined MM	none Machined MM	Straightness Flatness MM
<= 6	+ 0.1	+ 0.5	0.008
> 6 - 30	+ 0.2	+ 1.0	0.020
> 30 - 120	+ 0.3	+ 1.5	0.040
> 120 - 400	+ 0.5	+ 2.0	0.080
> 400 - 1000	+ 0.8	+ 3.0	0.150
> 1000 - 2000	+ 1.2	+ 4.0	0.200
> 2000 - 4000	+ 2.0	+ 6.0	
> 4000	+ 3.0	+ 8.0	

CONCENTRIC and AXIAL RUN TOLERANCE 0.1 MM

SYMMETRIC TOLERANCE 0.5 MM

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A02	21-07-20	ISSUED FOR APPROVAL	LG	RM	SC	RL	
		A01	03-06-20	ISSUED FOR APPROVAL	LG	RM	SC	RL	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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OUTOTEC DRAWING NUMBER: 900707 _GAU01 _AA04_MTA01 _38902 — DOCUMENT ID: OU100662224 — REVISION: 1

DRAWN	LokGav	21-07-20
CHECKED	Rohmul	21-07-20
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APPROVED	RicLeh	21-07-20
EMPLOYER		
SCALE	1:30	A1

MANSOURAH-MASSARAH GOLD PROJECT
OUTLINE DRAWING
SMBS & PER. STORG. & MIXING STN
3890-TNK-001

DRAWING NUMBER: MD-216-3890-EG-ME-GAD-1009 — REV: A02